

● PROBLEM STATEMENT

Pricing a home fairly is harder than most buyers and sellers assume — decisions are often anchored to gut feel or loose comparisons rather than the features that actually move price. This project applies machine learning to a 545-property housing dataset to predict sale price from structural and amenity attributes such as plot area, bathroom count, and air conditioning, with the goal of replacing intuition with a data-backed estimate.

● METHODOLOGY

Data Preparation

The dataset arrived clean — no missing values or duplicates. The single categorical field, *furnishing status* (furnished / semi-furnished / unfurnished), was one-hot encoded and cast to integer flags so both models could consume it directly.

Model Training

- **Linear Regression** — a weighted linear combination of all features.
- **Random Forest** — an ensemble of 200 trees built to capture non-linear interactions between features.

Data was split 80/20 into training and test sets, with 5-fold cross-validation used to confirm results held up beyond a single split.

● RESULTS

Model	MAE (Rs.)	RMSE (Rs.)	R²
Linear Regression	9,42,000	12,85,000	0.662
Random Forest	9,87,000	13,42,000	0.624

Linear Regression won on every metric. With only 545 rows to learn from, the Random Forest's 200 trees fit the training data too closely and lose some accuracy on unseen examples — a textbook case of overfitting on a small dataset.

● KEY FINDINGS

#1

AREA DOMINATES

Plot area is by far the strongest price driver, followed by bathroom count, air conditioning, and furnishing status. Guestrooms and hot-water heating barely register.

66%

VARIANCE EXPLAINED

The linear model accounts for roughly two-thirds of price variation. The remainder likely reflects factors absent from the data — exact location, property age, market timing.

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SIMPLICITY WINS

Random Forest looked like the safer bet for handling non-linearity, but price here scales cleanly with a handful of features — leaving little hidden complexity for it to exploit.

● BUSINESS RECOMMENDATION

Get the fundamentals right first. Since plot area and bathroom count carry the most predictive weight, accurate data collection on these two fields matters more than any modeling choice. On the product side, surfacing them as primary search filters — rather than secondary attributes — would give buyers the clearest read on what a property is really worth.